

Metric-Consistent Spectral Finite Element Decomposition of Stable Boundary Layer Regimes: A Riemannian Manifold Approach to Sub-Mesoscale Wave–Turbulence Partitioning

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Abstract

Nocturnal stable boundary layers (SBLs) exhibit complex interactions between three dominant scales: large-scale mesoscale pressure forcing, horizontally propagating internal gravity waves (IGWs), and highly localized turbulent shear bursts. Traditional diagnostic frameworks conflate these regimes, producing high variance in inversion-layer flux estimates and inconsistent model-observation comparisons. We present a *metric-consistent Riemannian pseudospectral* framework that projects sparse tower observations onto a non-uniform computational grid designed explicitly to maximize sensitivity near the surface (2.85 mm effective node resolution at $z = 1.5$ m) while maintaining numerical stability at upper boundaries ($\kappa(\mathbf{M}) = 4619$). Using Chebyshev polynomials as the spectral basis and solving the rank-deficient least-squares problem via thin singular value decomposition (SVD), we extract three physically meaningful diagnostic features: effective modal dimension D_{eff} , wave energy fraction F_W , and spectral curvature χ_N .

Applying this framework to the CASES-99 campaign stable window (22–31 October 1999), we identify *three statistically separable regimes* with a silhouette clustering score of $\bar{S} = 0.582$ for $K = 3$: (1) Continuous Turbulence ($D_{\text{eff}} > 20$, $F_W < 0.15$), (2) Wave-Dominated ($D_{\text{eff}} \sim 4\text{--}6$, $F_W > 0.75$), and (3) Intermittent Shear Bursts ($2 < D_{\text{eff}} < 15$, $0.15 < F_W < 0.50$). Crucially, the regime-specific rank compression ($\Delta_{\text{rank}} = 24$) proves that atmospheric dynamics naturally occupy lower-dimensional manifolds and are not mathematical artifacts of basis truncation.

We extend the theoretical framework by establishing an explicit physical link to the Ozmidov buoyancy scale L_O , resolving the grid-topology mismatch via a generalized metric-consistent Virtual Tower interpolation operator ($\mathcal{P}_{\text{VT}}$) for direct NWP model validation, and formulating a 2/3-rule spectral anti-aliasing filter for forward-predictive implementations. A synthetic gravity-wave reflection test confirms that the spectral partition functions suppress upper-boundary reflection by approximately 3.3×10^4 , validating the sponge-layer boundary conditions.

Keywords: stable boundary layers, spectral methods, manifold learning, gravity waves, Ozmidov scale, virtual tower interpolation

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